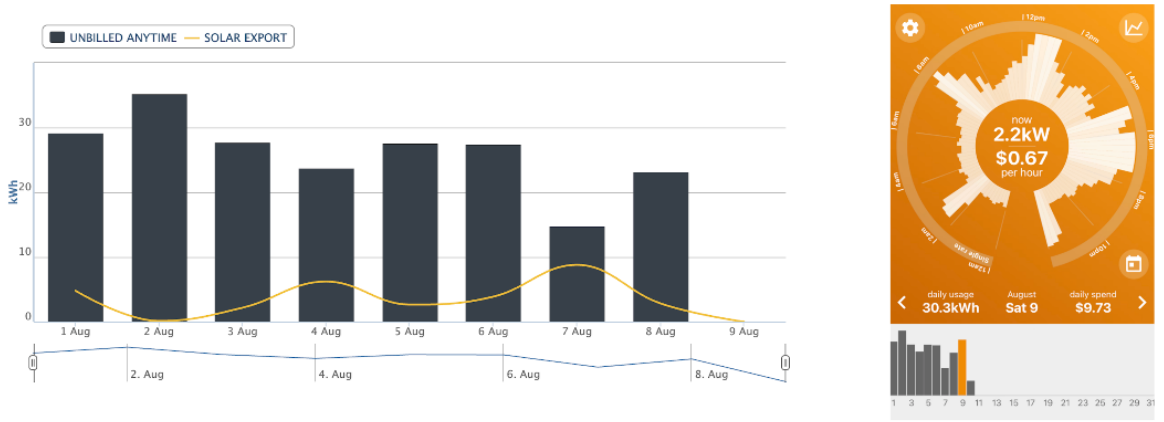
EXAMPLE: ELECTRICITY MONITORING IN THE HOME

Fitting my home out with automated electrical monitoring on every circuit is prohibitively expensive for the objectives of my home digital twin so I have fitted an optical pulse sensor to my electricity meter that records much finer grained data but only transfers the data to my phone. To ingest this data into my digital twin, I must export the data from my phone as a CSV file, and import it to my digital twin. Figure 3.8 shows an example of daily consumption data available to me from my electricity utility via a web portal compared to the data that is available from the new sensor, transmitted to my phone. Notice that my provider portal data is delayed by two days, making it impossible to make meaningful data driven decisions.



Current consumption data

Future state with granular data

Figure 3.8 The electricity consumption data that I currently have available from my provider on the left, compared with the fine grained data I will obtain with an optical sensor, right.

3.4.3 Manual sensing strategies

Manual data collection becomes valuable when automated sensing is impractical, too expensive, or when human observation adds important context. The key to successful manual sensing is to make manual collection systematic and integrated with your digital twin.